Ana III Zusammenfassung

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1 Measure Theory

Definition 1.1 (outer measure)**.** Let X be any set and let $\mu : 2^X \to [0, \infty]$ then μ is an outer measure if

1. $\mu(\emptyset) = 0$
2. $\mu(A) \leq \sum_k \mu(A_k)$ for every $A \subset \bigcup_k A_k$

Definition 1.2 (measurability)**.** A set $A \subset X$ is considered μ-measurable if for every $B \subset X$ we have the inequality

$$\mu(B) \geq \mu(A \cap B) + \mu(B \backslash A)$$

Now let $\mathcal{M}(\mu)$ be the set of all μ-measurable sets.

Theorem 1.3 (basic properties of measurable sets)**.** *Let $A \subset X$ be measurable and let $A_k \subset X$ be a sequence of measurable sets then we have the following properties*

1. $X \setminus A = A^C \in \mathcal{M}(\mu)$
2. $\bigcap_k A_k \in \mathcal{M}(\mu)$ *and* $\bigcup_k A_k \in \mathcal{M}(\mu)$
3. *if A_k is a pairwise disjoint sequence then we have $\mu(\bigcup_k A_k) = \sum_k \mu(A_k)$*

Proof. **1.)**
follows by realizing that the definitoin of measurability is invariant under complements ($B \setminus A = B \cap A^C$).
3.)
Using the fact that

$$A \cup B = (A \cap B) \cup (A \cap B^C) \cup (B \cap A^C)$$

and applying the definition of measurability twice we get that

$$\mu(E) \geq \mu(E \cap (A \cup B)) + \mu(E \cap (A \cup B)^C)$$

and thus $A \cup B$ is measurable now by using and inductive argument we get that every finite union of measurable sets is still measurable. Now define $B_N := \bigcup_{k=1}^N A_k$ then by measurability and induction we get that

$$\mu(B_{N+1}) = \mu(B_{N+1} \setminus A_{N+1}) + \mu(B_{N+1} \cap A_{N+1}) = \mu(B_N) + \mu(A_{N+1}) = \sum_{k=1}^{N+1} \mu(A_{N+1})$$

now by taking limits we get claim 3.
2.)
Now for 2 we define

$$\hat{A}_{N+1} = A_{N+1} \setminus (\bigcup_k^N A_k), \quad B_N = \bigcup_k^N \hat{A}_k$$

then we have that the $\hat{A}_k$ are disjoint and also

$$A = \bigcup_k^\infty A_k = \bigcup_k^\infty \hat{A}_k$$

Now we see by using 3 on the disjoint family we get

$$\mu(E) \geq \mu(E \cap B_N) + \mu(E \cap B_N^C) \geq \mu(E \cap B_N) + \mu(E \cap A^C)$$

Now by applying 3 we see that

$$\mu(E \cap B_N) = \sum_k^N \mu(E \cap \hat{A}_k)$$

by passing to the limit we get that

$$\mu(E) \geq \sum_k^\infty \mu(E \cap \hat{A}_k) + \mu(E \cap A^C) \geq \mu(E \cap A) + \mu(E \cap A^C)$$

which establishes claim 2. □

Definition 1.4 (σ-finite)**.** A measure space $(X, \mathcal{M}, \mu)$ is σ-finite if there are sets $E_n \in \mathcal{M}$ such that

$$X = \bigcup_n E_n, \quad \mu(E_n) < \infty$$

1.1 Lebesgue Measure

Definition 1.5 (rectangle)**.** Let $a, b \in \mathbb{R}^n$ such that $a_i \leq b_i$ then we define the rectangle with corners a and b by

$$Q(a, b) := (a_1, b_1) \times \cdots \times (a_n, b_n)$$

and we define the volume with

$$|Q(a, b)| := \prod_{i=1}^n (b_i - a_i)$$

Definition 1.6 (lebesgue outer measure)**.** Let $E \subset \mathbb{R}^n$ then we define the n dimensional Lebesgue Outer Measure by

$$\mathcal{L}^n(E) := \inf\{\sum_k |Q_k| : E \subset \bigcup_k Q_k, \text{ where } Q_k \text{ are rectangles}\}$$

Theorem 1.7 (properties of the lebesgue measure)**.** *The Lebesgue measure has the following properties*

1. $\mathcal{L}^n$ *is Borel regular and* σ*-finite*
2. $\mathcal{L}^n(Q(0,1)) = 1$
3. $\mathcal{L}^n(A + t) = \mathcal{L}^n(A)$ *where* $t \in \mathbb{R}^n$
4. $\mathcal{L}^n(\Lambda(A)) = \det(\Lambda)\mathcal{L}^n(A)$ *with* $\Lambda = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ *and* $\lambda_k \geq 0$ *for all* k

Proof. will be added later □

Optional

Note: In this section i tried to build a different proof of property 2 of the Lebesgue measure wich i find to be more intuitive to understand. So this section is not relevant to the lecture but might help to get a better picture of the intuition used in the proofs.

So we will first define a simpler way to measure a class of sets called elementary wich are sets that can be divided into rectangles. A set $Q \subset \mathbb{R}^n$ is called **elementary** if it can be represented as a finite union of n-dimensional rectangles with disjoint interiors.

Definition 1.8. Let $Q \subset \mathbb{R}^n$ be elementary with $Q = \bigcup_{i=1}^{N} Q_i$ and we will define

$$v(Q) := \sum_{i=1}^{N} |Q_i|$$

As a fun fact this measure is closely related to the Lebesgue measure in somewhat the same way as the Riemann integral is related to the Lebesgue integral. The Lebesgue version "*completes*" the Jordan measure/Riemann integral. The first observation that we will make about the Jordan measure is that is is well defined

Proposition 1.9 (well definedness)**.** *If $Q = \bigcup_{k=1}^{N} A_k = \bigcup_{k=1}^{M} B_k$ are two different representations then*

$$\sum_{k=1}^{N} |A_k| = \sum_{k=1}^{M} |B_k|$$

in other words v is well defined and does not depend on the representation.

Proof. Each rectangle A_k and B_j is the Cartesian product of n intervals. If for each coordinate axis $1 \leq i \leq n$, we collect all the endpoints of the one dimensional intervals that make up the rectangles A_k and B_j and sort them we get a finite Partition and we will call the closed non overlapping intervalls of this Partition P_i. Formally we have that $A_k = \prod_{i=1}^{n} [a_i, b_i]$ and $B_j = \prod_{i=1}^{n} [\alpha_i, \beta_i]$ (or any other way of invervalls i.e. open or half open). Thus the partition is the collecion of the endpoints of these intervals making up the Rectangles. Now we will define the grid generated by the P_i's by

$$G := \left\{ \prod_{i=1}^{n} I_i \quad : I_i \in P_i \right\} = (C_k)_{k=1}^{|P|}$$

Where C_k are the cells inside of the grid G. Now by the construction of the grid we see that each A_k can be expressed as a unique union of cells in this grid thus

$$A_k = \bigcup_{i \in I_k} C_i$$

which also works for B_k and thus we get

$$B_k = \bigcup_{i \in J_k} C_i$$

where the C_i's have disjoint interiors. Now because these cells form a perfect Partition of A_k we see that

$$|A_k| = \sum_{i \in I_k} |C_i|$$

The same works for B_k and we get that

$$|B_k| = \sum_{i \in J_k} |C_i|$$

Because the A_k's have disjoint interior we see that one cell can only be contained in one of the A_k's and thus the I_k's are pairwise disjoint this argument also works for the B_k. Now because they both make up the same rectangle Q we see that the Grid Cells must be the same and thus

$$\bigcup_{k=1}^{N} I_k = \bigcup_{k=1}^{M} J_k = S$$

Which instantly gives

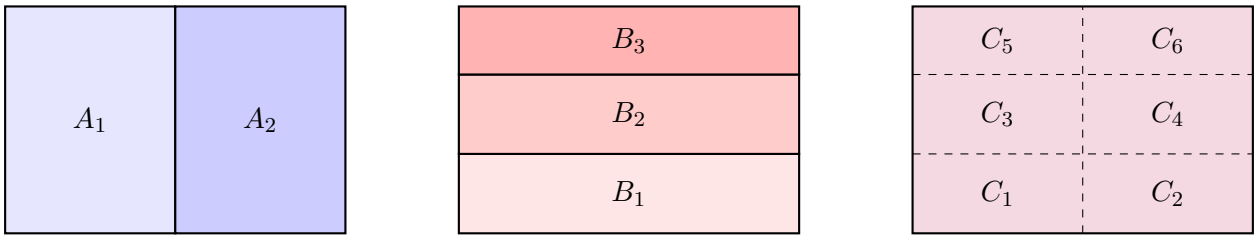
$$\sum_{k=1}^{N} |A_k| = \sum_{k=1}^{N} \sum_{i \in I_k} |C_i| = \sum_{i \in S} |C_i|$$

and

$$\sum_{k=1}^{M} |B_k| = \sum_{k=1}^{M} \sum_{i \in J_k} |C_i| = \sum_{i \in S} |C_i|$$

thus they are equal. □

Here is a quick Visualization trying to give a better understanding of the Stategy used in this proof as it is quite fundemental.



Representation A Representation B Common Grid G

Figure 1: The grid is formed by taking the Cartesian product of the partitions P_i from the axes, creating a common refinement $\{C_k\}$.

Now this Grid strategy is really useful and we will use it to proof the following proposition which will then be really helpfull in proving the claim.

Proposition 1.10. *Let Q be a rectangle and let $Q \subset \bigcup_{k=1}^{N} A_k$ where A_k are Rectangles then $|Q| \leq \sum_{k=1}^{N} |A_k|$*

Proof. Now let P_i be the partition generated by the one dimensional intervals of the A_k rectangles and Q and let $G = (C_i)_{i=1}^{|G|}$ be the Grid generated by those partitions. And now we see that each A_k can be written as some finite union of Grid cells

$$A_k = \bigcup_{i \in I_k} C_i$$

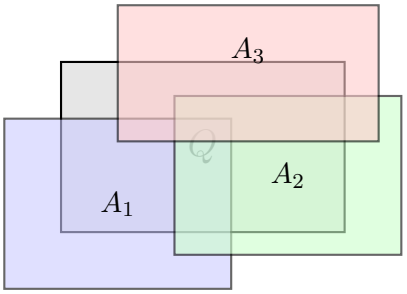
and we also get that

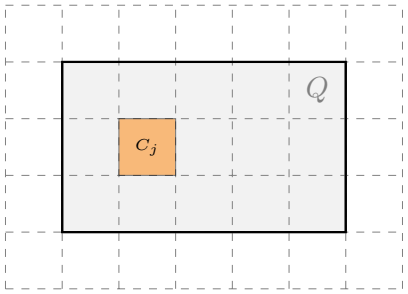
$$Q = \bigcup_{j \in J} C_j$$

Now by the above proposition we get that $|Q| = v(Q) = \sum_{j \in J} |C_j|$ and because the Cells in the grid are non overlapping we get that every cell that is inside Q but be inside some A_k and thus

$$\sum_{j \in J} |C_j| \leq \sum_{k=1}^{N} \sum_{i \in I_k} |C_i| = \sum_{k=1}^{N} v(A_k) = \sum_{k=1}^{N} |A_k|$$

□





Grid extending boundaries of Q and A_k. Every grid cell inside Q belongs to at least one A_k.

Figure 2: Subadditivity proof via grid technique: Q is fully covered by the overlapping rectangles A_k. The grid partitions the space so each cell in Q can be matched to a cell in at least one A_k.

Proposition 1.11. $\mathcal{L}^n(Q(0,1)) = 1$

. We easily see that $1 \geq \mathcal{L}^n(Q(0,1))$ now we let $\varepsilon > 0$ and we get that $Q_\varepsilon = [\varepsilon, 1-\varepsilon]^n \subset Q(0,1)$ if we let Q_k be any rectangles such that $Q(0,1) \subset \bigcup_{k=1}^{\infty} Q_k$ now because Q_ε is compact we get that there exist an N such that

$$Q_\varepsilon \subset \bigcup_{k=1}^{N} Q_k$$

now by using the prior proposition we get that

$$|Q_\varepsilon| \leq \sum_{k=1}^{N} |Q_k| \leq \sum_{k=1}^{\infty} |Q_k|$$

now by taking the inf over all the covers we get that

$$\mathcal{L}^n(Q(0,1)) \geq |Q_\varepsilon| = (1-2\varepsilon)^n$$

now by taking limit with $\varepsilon \to 0$ we get that $\mathcal{L}^n(Q(0,1) \geq 1$ which finishes the proof $\square$

1.2 Abstract Measure Spaces

Definition 1.12 (measure space)**.** A triple $(X, \mathcal{M}, \mu)$ is called a measure space if $\mu : \mathcal{M} \to [0, \infty]$ is a measure and $\mathcal{M}$ is a σ-Algebra.

Theorem 1.13 (monotone convergence for measures)**.** *Let $(A_n)_n$ be a sequence of measurable sets*

1. *if $A_1 \subset A_2 \subset \dots$ then we have*

$$\mu(\bigcup_n A_n) = \lim_{n\to\infty} \mu(A_n)$$

2. *if $A_1 \supset A_2 \supset \dots$ and $\mu(A_1) < \infty$ then we have*

$$\mu(\bigcap_n A_n) = \lim_{n\to\infty} \mu(A_n)$$

Proof. will be added later □

Definition 1.14 (Borel σ-Algebra)**.** If (X, d) is a metric space then we define the Borel σ-Algebra $\mathcal{B}(X)$ to be the smallest σ-Algebra containing all open sets of X meaning it is the intersection of all σ-Algebras that contain the open sets of X.

Definition 1.15 (Borel Regular)**.** A measure μ is Borel Regular if $\mathcal{B}(X) \subset \mathcal{M}(\mu)$

Proposition 1.16. *A measure μ is Borel Regular iff for every $A, B \in 2^X$ with* $\operatorname{dist}(A, B) > 0$ *we have*

$$\mu(A \cup B) = \mu(A) + \mu(B)$$